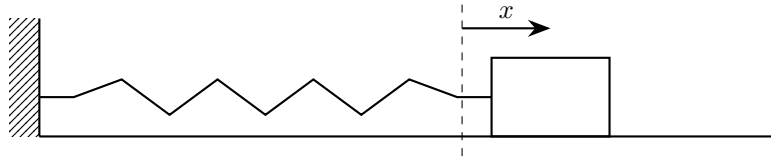


## Questions

|             |                 |    |
|-------------|-----------------|----|
| Question 1  | <i>14 marks</i> | 2  |
| Question 2  | <i>23 marks</i> | 3  |
| Question 3  | <i>13 marks</i> | 4  |
| Question 4  | <i>5 marks</i>  | 5  |
| Question 5  | <i>22 marks</i> | 6  |
| Question 6  | <i>14 marks</i> | 7  |
| Question 7  | <i>13 marks</i> | 8  |
| Question 8  | <i>12 marks</i> | 9  |
| Question 9  | <i>7 marks</i>  | 10 |
| Question 10 | <i>14 marks</i> | 11 |
| Question 11 | <i>12 marks</i> | 12 |
| Question 12 | <i>11 marks</i> | 13 |



1. The horizontal displacement of a glider attached to a spring, shown in the diagram above, is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 16x = 3 \sin 4t$$

where  $x$  is the displacement, in metres, of the glider from its equilibrium position,  $t$  seconds after the motion begins.

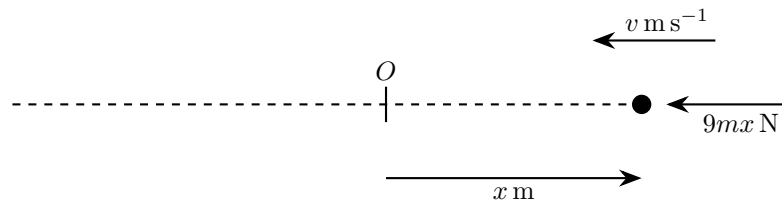
- (a) (i) Show that a particular solution of the differential equation is

$$x = -\frac{3}{8}t \cos 4t \quad [4]$$

- (ii) Hence, find the general solution of the differential equation [4]

Initially, the glider is 0.20 m to the right of equilibrium and is moving towards equilibrium at  $0.50 \text{ m s}^{-1}$ . Given that 4 seconds after the motion begins, the displacement of the glider is  $\beta$  m,

- (b) determine, according to the model, the value of  $\beta$  to 3 significant figures [4]
- (c) Given that the true value of  $\beta$  is 0.46 m, comment on the suitability of the model [1]
- (d) Write down a differential equation for the glider if the periodic driving force is removed, so that the motion becomes simple harmonic [1]



2. A particle of mass  $m$  kg moves in a horizontal line through its equilibrium position  $O$ . At time  $t$  seconds its displacement from  $O$  is  $x$  metres and its speed is  $v$   $\text{m s}^{-1}$ . When the particle is to the right of  $O$ , it is acted on by a restoring force of magnitude  $9mx$  newtons directed towards  $O$ , as shown in the diagram.

At  $t = 0$ , the particle is 1 m to the right of  $O$  and is projected towards  $O$  with speed  $6$   $\text{m s}^{-1}$ .

- (a) In the absence of damping and any external force,

- (i) Show that the motion is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 9x = 0 \quad [1]$$

- (ii) State the type of motion [1]  
 (iii) Write down the period of the motion [1]  
 (iv) Find  $x$  in terms of  $t$  [4]  
 (v) Find the amplitude of the motion [2]

- (b) In a second model, the particle again starts from the same initial position and velocity, but the motion is damped by a force  $6mv$  newtons acting opposite to the direction of motion.

- (i) Show that

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0 \quad [1]$$

- (ii) State, giving a reason, whether the system is under-damped, critically damped or over-damped [1]  
 (iii) Determine the general solution of this differential equation [3]

- (c) In a third model, the particle again starts from the same initial position and velocity, and is subject both to the damping force in part (b) and to an external force  $25m \cos 4t$  newtons acting in the positive direction. The motion is now modelled by the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 25 \cos 4t$$

- (i) Find  $x$  in terms of  $t$  [7]

- (ii) In the long term, the particle is observed to perform simple harmonic motion with period about 1.6 seconds. Verify that this is consistent with the answer to part (c)(i) [2]

- 3.** The charge  $q$  on a capacitor in a driven RLC circuit is modelled by

$$\frac{d^2q}{dt^2} + 2\frac{dq}{dt} + 10q = 37 \cos 3t$$

where  $t \geq 0$  is time.

When  $t = 0$ ,  $q = 1$  and  $\frac{dq}{dt} = 15$ .

**(a)** Find  $q$  as a function of  $t$  [9]

**(b)** Hence write down an approximate expression for  $q$  when  $t$  is large, giving your answer in the form  $R \cos(3t - \alpha)$ , where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$  [4]

4. The vertical displacement of the body of a car from its steady running height is measured by  $y$  metres, where  $y$  is positive upwards. The body can move above or below this level, so  $y$  can take positive and negative values. After the car leaves a speed hump, the body is momentarily above its steady running height and at rest at time  $t = 0$ .

A proposed model for the subsequent motion, for  $t \geq 0$ , is

$$4 \frac{d^2 y}{dt^2} + p \frac{dy}{dt} + 9y = 0,$$

where  $p$  is a constant and  $p \geq 0$ .

- (a) (i) According to the model, for what value of  $p$  would the motion be simple harmonic? [1]  
(ii) Explain briefly why modelling the motion as simple harmonic is unlikely to be realistic for a car suspension [1]
- (b) Find the range of values of  $p$  for which the model predicts that the body of the car never goes below its steady running height [2]
- (c) Sketch a possible graph of  $y$  against  $t$  when  $p$  lies outside the range found in part (b) but the motion is not simple harmonic [1]

5. A load of mass  $m$  kg hangs from a light vertical spring and is free to move only vertically in a liquid.

The spring has spring constant  $k \text{ N m}^{-1}$ , and the liquid exerts a resistance force of magnitude  $cv$  newtons opposite to the direction of motion, where  $v \text{ m s}^{-1}$  is the speed of the load.

Let  $y$  metres be the downward displacement of the load from its equilibrium position at time  $t$  seconds. Assume that  $4km > c^2$ .

- (a) Show that the motion is modelled by

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0$$

and hence show that

$$y = e^{-\frac{ct}{2m}} \left( P \cos \left( \frac{\sqrt{4km - c^2}}{2m} t \right) + Q \sin \left( \frac{\sqrt{4km - c^2}}{2m} t \right) \right), \quad [7]$$

where  $P$  and  $Q$  are constants.

- (b) It is given that

$$\begin{aligned} m &= 5 \\ c &= 30 \\ k &= 125 \end{aligned}$$

At time  $t = 0$ , the load is 0.12 m below its equilibrium position and is moving upwards with speed  $0.12 \text{ m s}^{-1}$ .

- (i) Find the values of  $P$  and  $Q$  [4]

- (ii) Find the first positive value of  $t$  for which the load is at its equilibrium position [5]

- (c) A second model ignores the resistance force but keeps the same values of  $m$  and  $k$ . Using the same initial conditions as in part (b), find an expression for  $y$  in terms of  $t$ , giving numerical constants to two significant figures where appropriate [6]

6. A manufacturer is testing the response of an electronic weighing platform. The displayed mass,  $m$ , measured in kilograms, after a box is placed on the platform is modelled by the differential equation

$$4 \frac{d^2m}{dt^2} + 4 \frac{dm}{dt} + 5m = 20$$

where  $t$  is the number of seconds after the box is placed on the platform.

- (a) Find a general solution of the differential equation [4]

Initially, the display reads 0 kg and the displayed mass is increasing at  $5 \text{ kg s}^{-1}$ .

- (b) Find, according to the model, the greatest value shown on the display [8]

The machine is programmed to print a label only when the displayed mass is less than 4.05 kg.

- (c) Determine whether, according to the model, the machine will print a label exactly 8 seconds after the box is placed on the platform [2]

7. An analogue meter has a damped pointer. After a short electrical pulse, the angular displacement  $\theta$  radians of the pointer from its rest position, measured positive clockwise, is modelled by

$$\frac{d^2\theta}{dt^2} + 5\frac{d\theta}{dt} + 6\theta = 4e^{-4t}$$

where  $t$  is the time, in seconds, after the motion begins.

In one test, at  $t = 0$  the pointer is passing through its rest position and has angular velocity  $1 \text{ rad s}^{-1}$  anticlockwise.

For this test, use the model to

- (a) find an equation for  $\theta$  in terms of  $t$  [8]

- (b) find, to 3 decimal places, the greatest clockwise angular displacement of the pointer from its rest position [4]

In this test, the time taken for the pointer to pass through its rest position again was measured as 0.69 seconds.

- (c) State, with justification, whether or not this supports the model [1]

8. A trolley moves on a horizontal track and is attached to a spring and a dashpot. The trolley is pulled to the right of its equilibrium position and released.

At time  $t$  seconds after release, its horizontal displacement is  $x$  metres to the right of a fixed point  $P$ , where  $P$  is the equilibrium position and is 1.2 metres from the left-hand end of the track.

When  $t = 0$ :

$$\begin{aligned}x &= 0.18, \\ \text{the velocity of the trolley is } &0 \text{ m s}^{-1}, \\ \text{the acceleration of the trolley is } &-0.10 \text{ m s}^{-2}\end{aligned}$$

In the subsequent motion, the displacement of the trolley is modelled by the differential equation

$$3k \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5x = 0, \quad t \geq 0,$$

where  $k$  is a positive constant.

- (a) Find the value of  $k$  [2]
- (b) Find the particular solution of the differential equation [7]
- (c) Hence find, according to the model, the distance of the trolley from the left-hand end of the track 4.5 seconds after it is released [2]
- (d) State one limitation of the model [1]

**9.** In a feedback system, the error signal  $E$  is measured in suitable units.

For  $t \geq 0$ ,  $E$  is modelled by the differential equation

$$\frac{d^2E}{dt^2} + \frac{dE}{dt} - 6E = -12$$

Initially the error is decreasing at 9 units per second, so  $\frac{dE}{dt} = -9$  when  $t = 0$ . It is given that  $E$  tends to a finite limit as  $t \rightarrow \infty$ .

(a) Determine the value of  $E$  when  $t = \ln 2$  [6]

(b) Determine the limit of  $E$  as  $t \rightarrow \infty$  [1]

- 10.** A machine part executes damped forced vibrations. Its displacement  $y$  centimetres from equilibrium at time  $t$  seconds is modelled by

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 10y = 20\cos t + 5\sin t, \quad t \geq 0$$

- (a) Find the general solution of this differential equation. [6]
- (b) Given that  $y = 3$  and  $\frac{dy}{dt} = 3$  when  $t = 0$ , find the particular solution describing the motion. [4]
- (c) The first turning point of this motion for  $t > 0$  occurs when  $y = a$ . Find  $a$  correct to 3 significant figures. [4]

11. A particle  $Q$  moves along the  $x$ -axis under a single force. When the time is  $t$  seconds, the displacement of  $Q$  from the origin is  $x$  metres and its velocity is  $v \text{ m s}^{-1}$ .

The motion of  $Q$  is modelled by the differential equation

$$\ddot{x} + \lambda^2 x = 0$$

where  $\lambda \text{ rad s}^{-1}$  is a positive constant.

At time  $t = 0$ ,  $Q$  is at the point  $A$  where  $x = -\frac{1}{2}a$ , for some positive constant  $a$ , and is moving in the positive direction.

Later,  $Q$  reaches the point  $B$  where  $x = a$  and comes to instantaneous rest.

- (a) Write down the general solution of this differential equation. [1]

- (b) Using the answer to part (a), determine expressions, in terms of  $\lambda$ ,  $a$  and  $t$  only, for the following quantities:

- (i)  $x$  [3]  
(ii)  $v$

- (c) Hence show that

$$v^2 = \lambda^2(a^2 - x^2) \quad [1]$$

- (d) The quantity  $r$  is defined by  $r = \frac{1}{v}$ . Using part (c), determine an expression for  $r_m$ , the mean value of  $r$  with respect to the displacement, as  $Q$  moves directly from  $A$  to  $B$ . [4]

- (e) One measure of the validity of the model is consideration of the value of  $r_m$ . If  $r_m$  exceeds 7.2 then the model is considered to be valid. The value of  $a$  is measured as 0.240 to 3 significant figures. The value of  $\lambda$  is measured as  $0.80 \pm 0.02$ . Determine what can be inferred about the validity of the model from the given information. [1]

- (f) Find, according to the model, the least possible value of the velocity of  $Q$  at time  $t = 0$ . Give your answer to 2 significant figures. [2]

- 12.** An automated inspection trolley of mass 2 kg moves along a straight horizontal rail. The point  $A$  is fixed on the rail. The displacement of the trolley from  $A$  at time  $t$  seconds is  $x$  metres, for  $t \geq 0$ .

For the interval considered, the trolley stays to the right of  $A$  and continues to move away from  $A$ .

For  $t \geq 0$ , the trolley is subject to three forces modelled as follows:

- A driving force of magnitude  $4(t + 1)$  N acts in the positive  $x$ -direction.
- A magnetic force of magnitude  $x$  N acts towards  $A$ .
- A resistive force of magnitude  $2 \left| \frac{dx}{dt} \right|$  N acts opposite to the direction of motion.

- (a) Applying Newton's second law gives the differential equation

$$4(t + 1) - x - 2 \frac{dx}{dt} = 2 \frac{d^2x}{dt^2}$$

- (i) Explain the sign of the term involving  $\frac{dx}{dt}$  in this differential equation [1]

- (ii) When  $t = 0$  the displacement of the trolley is 1 m, and it is moving away from  $A$  with speed  $1 \text{ m s}^{-1}$ . Without attempting to solve the differential equation, find the acceleration of the trolley when  $t = 0$  [2]

- (b) Let the particular solution to the differential equation in part (a) be a function  $g$  such that  $x = g(t)$  for  $t \geq 0$ . The particular solution can be expressed as a Maclaurin series.

- (i) Show that the Maclaurin series for  $g(t)$  up to and including the term in  $t$  is  $1 + t$  [1]

- (ii) Use your answer to part (a)(ii) to show that the term in  $t^2$  in the Maclaurin series for  $g(t)$  is  $\frac{1}{4}t^2$  [1]

- (iii) By differentiating the differential equation in part (a) with respect to  $t$ , show that the term in  $t^3$  in the Maclaurin series for  $g(t)$  is  $\frac{1}{6}t^3$  [4]

- (c) You are given that the complete Maclaurin series for the function  $g$  is valid for all  $t \geq 0$ . After 0.4 seconds a sensor records that the trolley is 1.45 m from  $A$ .

- (i) Using the Maclaurin series for  $g(t)$  up to and including the term in  $t^3$ , evaluate the suitability of the model for determining the displacement of the trolley from  $A$  when  $t = 0.4$  [1]

- (ii) Explain why it would not be sensible to use the Maclaurin series for  $g(t)$  up to and including the term in  $t^3$  to evaluate the suitability of the model for determining the displacement of the trolley from  $A$  when  $t = 8$  [1]